

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Duolin Inhaler (CFC Free)

Composition

Each actuation delivers
Salbutamol Sulphate BP
Equivalent to Salbutamol100 mcg
Ipratropium Bromide monohydrate BP equivalent
to Ipratropium Bromide anhydrous20 mcg
Suspended in
Propellant HFA-227.....q.s

Description

A one piece filled metal container fitted with a valve with plastic stem and printed with product name, batch number and container number. On visual examination there should be no sign of physical damage such as dents, bulged containers bent valve stem and uneven crimping. A white coloured powder is obtained after opening the container.

Pharmacodynamics

Ipratropium bromide is an anticholinergic agent which inhibits vagally mediated reflexes by antagonising the action of acetylcholine, the transmitter agent released from the vagus nerve. The bronchodilation following inhalation of ipratropium bromide is primarily local and site specific to the lung and not systemic in nature.

Salbutamol sulphate is a β_2 -adrenergic agent which acts on airway smooth muscle resulting in relaxation. Salbutamol relaxes all smooth muscle from the trachea to the terminal bronchioles and protects against all bronchoconstrictor challenges.

The combination of ipratropium with salbutamol provides the simultaneous release of ipratropium bromide and salbutamol sulphate allowing the synergistic efficacy on the muscarinic and β_2 -adrenergic receptors in the lung to cause bronchodilation.

Pharmacokinetics

Ipratropium bromide is not readily absorbed into the systemic circulation either from the surface of the lung or from the gastrointestinal tract as compared by blood level and renal excretion studies. The half-life elimination of drug and metabolites is about 3 - 4 hours after inhalation or intravenous administration. Ipratropium bromide does not penetrate the blood brain barrier.

Salbutamol sulphate is rapidly and completely absorbed following oral administration either by the inhaled or gastric route. Peak plasma salbutamol concentrations are seen within three hours of administration and the drug is excreted unchanged in the urine after 24 hours. Intravenous salbutamol will cross the blood brain barrier reaching concentrations amounting to about five percent

of the plasma concentrations. From a pharmacokinetic perspective, the additive activity of the combination is due to the local effect of the fixed dose of the active components (ipratropium bromide and salbutamol sulphate) on the muscarinic and beta₂-adrenergic receptors in the lung.

Indications

Duolin is indicated for the management of reversible bronchospasm associated with obstructive airway diseases in patients who require more than a single bronchodilator

Dosage and Administration

Adults (including elderly patients): Two inhalations four times a day.

Children: There is no experience of the use of the combination of salbutamol and ipratropium in children below the age of 12 years.

Contraindications

Duolin is contraindicated in patients with hypertrophic obstructive cardiomyopathy or tachyarrhythmia, and in patients with a history of hypersensitivity to any of its components, or to atropine or its derivatives.

Duolin is also contraindicated in patients with a history of hypersensitivity to soya lecithin or related food products such as soya bean and peanut.

Warnings and Precautions

Special warnings and precautions for use immediate hypersensitivity reactions may occur after administration of Duolin, as demonstrated by rare cases of urticaria, angioedema, rash, bronchospasm and oropharyngeal oedema.

Ocular complications: There have been rare reports of ocular complications (i.e. mydriasis, blurring of vision, narrow-angle glaucoma, eye pain) when the contents of metered aerosols containing ipratropium bromide have been sprayed inadvertently into the eye. Care must be taken to prevent Duolin from entering the eye, particularly in patients who may be pre-disposed to glaucoma. Such patients should be specifically warned to protect their eyes.

Eye pain or discomfort, blurred vision, visual halos or colored images, in association with red eyes from conjunctival congestion and corneal oedema may be signs of acute narrow-angle glaucoma. Should any combination of these symptoms develop, treatment with miotic drops should be initiated and specialist advice sought immediately.

Patients must be instructed in the correct administration of Duolin Inhaler.

In the following conditions Duolin should only be used after careful risk/benefit assessment: Insufficiently controlled diabetes mellitus, recent myocardial infarction and/or severe organic heart or vascular disorders, hyperthyroidism, pheochromocytoma, prostatic hypertrophy and risk of narrow-angle glaucoma.

The patient should be instructed to consult a doctor immediately in the event of acute, rapidly worsening dyspnoea. In addition, the patient should be warned to seek medical advice should a reduced response become apparent.

Potentially serious hypokalemia may result from beta₂-agonist therapy. Particular caution is advised in severe asthma, as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids and diuretics. Additionally, hypoxia may aggravate the effects of hypokalemia on cardiac rhythm, especially in patients receiving digoxin. It is recommended that serum potassium levels are monitored in such situations.

Patients with cystic fibrosis may be more prone to gastro-intestinal motility disturbances.

Drug Interactions

Beta-adrenergics, xanthine derivatives and corticosteroids may enhance the effect of Duolin. The concurrent administration of other beta-mimetics, systemically absorbed anticholinergics and xanthine derivatives may increase the side effects.

A potentially serious reduction in effect may occur during concurrent administration of beta-blockers.

Beta-adrenergic agonists should be administered with caution in patients being treated with monoamine oxidase inhibitors or tricyclic antidepressants, since the action of beta-adrenergic agonists may be enhanced.

Inhalation of halogenated hydrocarbon anaesthetics such as halothane, trichloroethylene and enflurane may increase the susceptibility to the cardiovascular effects of beta-agonists.

Anticholinergic effects of other drugs can be enhanced.

Pregnancy and Lactation

Ipratropium bromide has been in general use for several years and there is no definite evidence of ill-consequence during pregnancy; animal studies have shown no hazard.

Salbutamol has been in widespread use for many years without apparent ill-consequence during pregnancy. There is inadequate published evidence of safety in the early stages of human pregnancy but in animal studies there has been evidence of some harmful effects on the foetus at very high dose levels.

As with all medicines, Duolin should not be used in pregnancy, especially the first trimester, unless the expected benefit is thought to outweigh any possible risk to the foetus. Similarly, Duolin should not be administered to breast-feeding mothers unless the expected benefit is thought to outweigh any possible risk to the neonate.

Side Effects

In common with other beta-agonists more frequent undesirable effects of Duolin are fine tremor of skeletal muscles and nervousness, less frequent are tachycardia, dizziness, palpitations or headache, especially in hypersensitive patients.

Potentially serious hypokalemia may result from beta₂-agonist therapy.

In isolated cases there may be local reactions such as dryness of the mouth, throat irritation, dysphonia or allergic reactions.

As with other bronchodilators, in some cases cough, in very rare instances paradoxical bronchoconstrictions have been observed.

As with other beta-mimetics, nausea, vomiting, sweating, weakness and myalgia/muscle cramps may occur. In rare cases decrease in diastolic blood pressure, increase in systolic blood pressure, arrhythmias, particularly after higher doses may occur.

In individual cases psychological alterations have been reported under inhalational therapy with beta-mimetics.

Isolated reports of ocular complications (i.e. mydriasis, increased intraocular pressure, angle-closure glaucoma, eye pain) when aerosolised ipratropium bromide either alone or in combination with an adrenergic beta₂-agonist has escaped into the eyes.

Ocular side effects, gastro-intestinal motility disturbances and urinary retention may occur in rare cases and are reversible.

Allergic-type reactions such as skin rash, pruritus, angioedema of the tongue, lips and face, urticaria (including giant urticaria), laryngospasm and anaphylactic reactions have been reported. Many of the patients have had a history of allergy to other drugs and/or foods, including soya bean.

The possibility of anaphylactic reactions to active or inactive constituents of Duolin is acknowledged.

As with all beta₂-agonists hyperactivity in children is possible.

Overdosage

The effects of overdosage are expected to be primarily related to salbutamol because acute overdosage with ipratropium bromide is unlikely as it is not well absorbed systemically after inhalation or oral administration. Hypokalaemia may occur following overdose. Serum potassium levels should be monitored.

Manifestations of overdosage with salbutamol may include anginal pain, hypertension, hypokalaemia and tachycardia. The preferred antidote for overdosage with salbutamol is a cardioselective beta-blocking agent but due care and attention should be used in administering these drugs in patients with a history of bronchospasm.

Storage:

Store below 30°C, Protect from frost.

Shelf life:

12 months

Packing/Pack size:

Duolin inhaler : 200 metered dose

Registration holder in Malaysia

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